

A Machine-Checked Renormalized Local Trace Formula at the Archimedean Place

Formalization in Lean 4 of the Connes–Consani Archimedean Weil Positivity
Programme,
with an Exact Evaluation of the Sonin Finite-Part Constant

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Abstract

We report a complete, **sorry**-free Lean 4 formalization of the operator-theoretic side of Connes and Consani’s programme on Weil positivity and the trace formula at the archimedean place, together with one new mathematical result obtained in the course of the verification.

Working on $L^2(\mathbb{R})$ with the sharp cut-offs $P^{(\Lambda)}$ (multiplication by $\mathbf{1}_{[-\Lambda, \Lambda]}$) and $\widehat{P}^{(\Lambda)} = \mathcal{F}^{-1} \mathbf{1}_{[-\Lambda, \Lambda]} \mathcal{F}$ and the Sonin sandwich $S^{(\Lambda)} = P^{(\Lambda)} \widehat{P}^{(\Lambda)} P^{(\Lambda)}$, we prove in Lean: (i) the semi-local trace density $\kappa_\Lambda(a) = \text{Tr}(D_a S^{(\Lambda)})$ in exact closed form in terms of the sine integral Si ; (ii) its off-diagonal limit $\kappa_\Lambda(a) \rightarrow \sqrt{a}/|1-a|$, the kernel of the archimedean Weil distribution; (iii) an unconditional logarithmic profile for a reference bump, hence an unconditional finite-part asymptotic on the full line with coefficient $4f(1) \log \Lambda$; and (iv), on the even (Sonin) subspace, the renormalized local trace formula

$$\text{Tr}(\vartheta(f) S^{(\Lambda)} P_{\text{ev}}) - 2f(1) \log \Lambda \rightarrow W_{\mathbb{R}}(\Delta^{-1/2} F) + f(1) z_{\text{ev}}, \quad F = \log\text{-transform of } f,$$

with the logarithmic coefficient exactly $2f(1)$, the paper’s normalization, and the odd-part remainder cancelling identically.

We further prove three sharp negative results, also machine-checked: the trace identity with the cut-off frozen at $\Lambda = 1$ is *false* for every unitary identification; the unnormalized Weil positivity statement is *false*, with an explicit counterexample; and the diagonal density satisfies $d(\Lambda) = 4\Lambda^2$ exactly, refuting the logarithmic diagonal hypothesis on which an earlier route depended.

The new mathematical contribution is an exact evaluation of the residual constant. We show that the sine-integral logarithm constant appearing in the formalization, $K = \int_0^1 \text{Si}(s) \frac{ds}{s} + \int_1^\infty (\text{Si}(s) - \frac{\pi}{2}) \frac{ds}{s}$, equals $\pi\gamma/2$, whence $z_{\text{ev}} = 2K/\pi - \gamma = 0$. The Sonin finite part is therefore *exactly* the normalized archimedean Weil functional, with no local correction term: $\text{Tr}(\vartheta(f) S^{(\Lambda)} P_{\text{ev}}) - 2f(1) \log \Lambda \rightarrow D(F) - L_{\text{Norm}}(F)$.

The development comprises 305 Lean modules and approximately 2.2×10^5 lines; every result reported here is free of **sorry** and depends only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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1 Introduction

1.1 Background

Weil’s explicit formula expresses a sum over the zeros of a Hecke L -function as a sum of local distributions, one for each place of the ground field [10, 1]. Weil’s positivity criterion states that the Riemann hypothesis is equivalent to the positivity of the resulting functional on a suitable class of test functions. In Connes’ spectral interpretation [2] the local terms arise as traces of cut-off scaling operators, and the archimedean place carries the analytically hardest contribution.

Connes and Consani [3] isolate the archimedean place and prove positivity there. Two objects are central to their treatment. The first is the *normalized* archimedean functional

$$L_{\text{Norm}}(F) = D(F) - W_{\mathbb{R}}\left(\Delta^{-1/2}F\right), \quad \Delta^{1/2}f(x) = x^{1/2}f(x),$$

where $W_{\mathbb{R}}$ is the archimedean Weil distribution and D pairs the test function against the trace remainder δ . The second is the *Sonin space*: the range of the projection $S = P\widehat{P}P$ built from a sharp cut-off in position and in frequency, whose spectral theory is that of the prolate spheroidal wave functions [8, 5].

The identity that carries the geometric content is a *local trace formula*: the trace of the scaling representation against the Sonin sandwich should reproduce the archimedean Weil term, once the logarithmic divergence produced by the cut-off is subtracted.

1.2 What is formalized here

This paper reports a Lean 4 development that carries this identity from an informal statement to a machine-checked theorem in a concrete model, and, in the course of doing so, settles several structural questions in the negative and evaluates the last remaining constant in closed form.

The three headline positive results are Theorems 4.2, 5.3 and 6.3; the novel closed-form evaluation is Theorem 7.2. Section 3 collects the negative results, which are of independent interest because each of them closes off a route that looks plausible from the informal statement.

We stress at the outset what is *not* claimed. The development proves a renormalized trace formula in a concrete model on $L^2(\mathbb{R})$; the identification of that model's trace functional with the abstract trace functional parameterized by an arbitrary unitary identification of the two pictures is not yet formalized. Section 10 states this precisely.

1.3 Authorship and method

The development was produced by a multi-agent workflow. The DULA programme supplied the mathematical targets, the model, and the research direction. Grok 4.5 carried out autonomous Lean 4 proof development across the runs recorded in the project log. Claude Opus 5 performed adversarial verification of each delivered artifact — checking statements against claims, auditing for vacuous hypotheses, and reproducing the analytic constants numerically from the Lean definitions — and contributed the closed-form evaluation of Section 7. We regard this division of labour as methodologically significant and return to it in Section 11.

2 The setting

2.1 The multiplicative group and its Haar measure

Let $\mathbb{R}_+^*$ denote the multiplicative group of positive reals with Haar measure $d^*\lambda$, and write $\lambda = e^u$ for the logarithmic coordinate, so that $d^*\lambda$ corresponds to Lebesgue measure du on $\mathbb{R}$. Test functions are elements of $C_c(\mathbb{R}_+^*, \mathbb{C})$; for $f \in C_c(\mathbb{R}_+^*, \mathbb{C})$ we write $F = \text{log-transform}(f)$, $F(u) = f(e^u)$, denoted `logTest f` in the formalization.

2.2 The scaling representation and the cut-offs

On $L^2(\mathbb{R})$ the dilation unitary is

$$(D_a \xi)(x) = \sqrt{a} \xi(ax), \quad a \in \mathbb{R}_+^*,$$

formalized in `Dilation.lean` together with its algebraic and metric API and the intertwining relation `weilMap_scaling_eq_dil` identifying it with the scaling representation $\vartheta(\lambda)$ of the paper: $w(\vartheta(\lambda)\xi) = D_{\lambda^{-1}}(w\xi)$.

The cut-offs are

$$P^{(\Lambda)} = \text{multiplication by } \mathbf{1}_{[-\Lambda, \Lambda]}, \quad \widehat{P}^{(\Lambda)} = \mathcal{F}^{-1} \mathbf{1}_{[-\Lambda, \Lambda]} \mathcal{F},$$

and the *Sonin sandwich* is

$$S^{(\Lambda)} = P^{(\Lambda)} \widehat{P}^{(\Lambda)} P^{(\Lambda)} = B_{\Lambda}^* B_{\Lambda}, \quad B_{\Lambda} = \widehat{P}^{(\Lambda)} P^{(\Lambda)}.$$

That B_{Λ} is Hilbert–Schmidt, that $S^{(\Lambda)}$ is positive, self-adjoint and trace class, and that $\vartheta(f)S^{(\Lambda)}$ is trace class with a basis-independent, absolutely convergent trace series, are all proved in the development (`CutoffFamily`, `CutoffFamilyHS`, `CutoffHilbertSchmidt`, `TraceClass`).

2.3 The semi-local density

Definition 2.1 (`SemiLocalKernel.lean`). The *semi-local trace density* is

$$\kappa_{\Lambda}(a) = \text{Tr}\left(D_a S^{(\Lambda)}\right),$$

computed along a Hilbert basis. In Lean:

```
def semiLocalDensity (b : HilbertBasis iota C L2R) (cut : R) (a : Rplus) : C :=
  traceAlong b (dilCLM a .. soninSandwichCut cut)
```

This is a genuine operator trace, not a closed form posited by fiat; the closed form is a theorem (Theorem 4.1 below). The corresponding trace functional in kernel form is

$$\text{Tr}\left(\vartheta(f)S^{(\Lambda)}\right) = \int_{\mathbb{R}_+^*} f(\lambda) \kappa_{\Lambda}(\lambda^{-1}) d^* \lambda,$$

proved as `cutTrace_eq_integral` by dominating the trace series uniformly in λ by $\|B_{\Lambda}\|_{\text{HS}}^2$ and interchanging summation with the Bochner integral.

3 Negative results: three routes closed

A distinctive feature of this development is that several plausible readings of the informal statement are not merely unproved but demonstrably false. Each refutation is machine-checked.

3.1 The fixed-cut-off trace identity is false

Theorem 3.1 (`TraceIdentityObstruction.lean`). For every unitary identification U of $L^2(\mathbb{R}_+^*, d^* \rho)$ with $L^2(\mathbb{R})$, the identity $L_{\text{Norm}}(f) = \text{Tr}(\vartheta(f)P_1 \widehat{P}_1 P_1)$ fails: $\neg \text{NormalizedTraceIdentity } U$.

The mechanism is a mismatch of orders. With the cut-off frozen at $\Lambda = 1$ the trace density κ is bounded, so $f \mapsto \int f \kappa d^* \lambda$ is continuous for the $L^1(d^* \lambda)$ norm; but L_{Norm} is a distribution of order one. Evaluating both sides on the triangular bump tri_w of half-width w , the closed form of the normalized archimedean term gives

$$\text{Re } L_{\text{Norm}}(\text{tri}_w) = \text{Re } D(\text{tri}_w) - c_0 + \int (1 - \text{tri}_w(u)) \kappa_{\infty}(u) du,$$

and the singularity $\kappa_{\infty}(u) \approx 1/(2|u|)$ of the explicit-formula kernel at the origin forces the last term to be at least $\frac{1}{6} \log(1/w)$, while $\text{Re } D(\text{tri}_w) \in [0, 16 \text{ Si } \pi + 16]$ and $\|\text{tri}_w\|_{L^1(d^* \lambda)} \leq 2w$. Letting $w \rightarrow 0$, the analytic side diverges and the trace side tends to 0.

Theorem 3.1 is not in conflict with [3]: it shows that the trace formula must be taken in its *renormalized* form, with $\Lambda \rightarrow \infty$ and a $\log \Lambda$ subtraction, which is exactly what the remainder of this paper establishes.

3.2 The unnormalized positivity statement is false

Theorem 3.2 (`UnnormalizedCounterexample.lean`). *Let $L = D + W_\infty$ pair both summands with f itself, rather than applying the archimedean distribution to $\Delta^{-1/2}f$. Then positivity fails: $\neg L\text{Positivity}$, witnessed by an explicit triangular (Fejér) bump.*

The obstruction is exact and is itself formalized: `WeilR_ofLog_eq_add_deficit` gives

$$W_{\mathbb{R}}(f) - W_{\mathbb{R}}(\Delta^{-1/2}f) = \int_0^\infty F(u) \frac{(e^{u/2} - 1)^2}{e^u - e^{-u}} du \geq 0,$$

a deficit growing with the support width while the normalized functional remains bounded. The normalized statement — Corollary 2.3(i) of [3] — is by contrast proved unconditionally in the development (`LPositivityNorm_holds`), for every C^4 compactly supported positive-definite test function, via the Fourier-side inequality $2\theta'(t) + \hat{\delta}(t) \geq 0$ established for all real t .

3.3 The diagonal density is quadratic, not logarithmic

Theorem 3.3 (`DiagonalExact.lean`, `DiagonalQuadraticGrowth.lean`). *The diagonal density satisfies exactly*

$$d(\Lambda) = \text{Tr}(S^{(\Lambda)}) = \|B_\Lambda\|_{\text{HS}}^2 = 4\Lambda^2, \quad \Lambda \geq 0.$$

Consequently the hypothesis $d(\Lambda) \leq 2 \log \Lambda + C$ fails for every constant C (`not_diagLogUpperBound`), and a fortiori $d(\Lambda) = 2 \log \Lambda + c + o(1)$ fails for every c .

The lower bound was first obtained by an orthonormal-family argument: by Jordan’s inequality a normalized box of width $1/(2\Lambda)$ retains at least $4/\pi^2$ of its energy in the frequency window $[-\Lambda, \Lambda]$, and the $[2\Lambda^2]$ disjoint translates of such a box inside $[-\Lambda, \Lambda]$ are orthonormal, giving $d(\Lambda) \geq (8/\pi^2)\Lambda^2 - 4/\pi^2$. The exact value $4\Lambda^2$ followed from a direct Hilbert–Schmidt computation.

This refutes the reduction of the logarithmic divergence to the diagonal value $d(\Lambda)$ — the route to which the programme had previously been narrowed. It does *not* refute the trace asymptotics itself: κ_Λ is a positive-definite function of λ reaching height $\Theta(\Lambda^2)$ in a window of width $\sim \Lambda^{-2}$ about the diagonal, so its *average* against a fixed test function can still be of size $\log \Lambda$. The correct route, developed below, computes the average directly.

4 The exact semi-local density and the off-diagonal limit

4.1 Closed form

Theorem 4.1 (`SemiLocalSi.lean`). *For $\Lambda \geq 0$ and $a \neq 1$,*

$$\kappa_\Lambda(a) = \sqrt{a} \frac{2}{\pi(a-1)} \text{Si}\left(\frac{2\pi(a-1)\Lambda^2}{\max(1, a)}\right).$$

Moreover the bound

$$|\kappa_\Lambda(a)| \leq \frac{2 \text{Si}(\pi)}{\pi} \cdot \frac{\sqrt{a}}{|a-1|}$$

holds uniformly in the cut-off.

The proof evaluates the double phase integral

$$\kappa_\Lambda(a) = \sqrt{a} \int_{|x| \leq \Lambda} \int_{|v| \leq \Lambda, |av| \leq \Lambda} e^{2\pi i(a-1)vx} dv dx$$

obtained from the factorization $S^{(\Lambda)} = G_\Lambda^* G_\Lambda$ and the Hilbert–Schmidt pairing identity `hsInner_eq_integral_0`. The consistency check $\kappa_\Lambda(1) = 4\Lambda^2$ recovers Theorem 3.3.

The cut-off-independent bound is the analytic engine of everything that follows: it supplies the dominating function for every subsequent dominated-convergence argument.

4.2 Recovery of the archimedean Weil kernel

Theorem 4.2 (`SemiLocalSi.lean`). *For every $a \neq 1$,*

$$\mathrm{Tr}\left(\vartheta(\lambda)S^{(\Lambda)}\right) \xrightarrow{\Lambda \rightarrow \infty} \frac{\lambda^{1/2}}{|1 - \lambda|},$$

equivalently, in the logarithmic coordinate $\lambda = e^u$ with $u \neq 0$,

$$\kappa_\Lambda(e^{-u}) \longrightarrow \frac{1}{2 \sinh(|u|/2)}.$$

This is exactly the kernel of the archimedean Weil distribution away from the diagonal. Since $\mathrm{Si}(x) \rightarrow \pi/2$ as $x \rightarrow +\infty$ and Si is odd, the limit follows from Theorem 4.1 with the sign of $a - 1$ absorbed into the absolute value.

4.3 The kernel comparison

Write, as in `ArchimedeanKernel.lean`,

$$\sinh\mathrm{Kernel}(u) = \frac{e^{|u|/2}}{e^{|u|} - e^{-|u|}}, \quad \sinh\mathrm{KernelRef}(u) = \frac{e^{-|u|/2}}{e^{|u|} - e^{-|u|}}.$$

The first is exactly the kernel of the archimedean Weil distribution $W_{\mathbb{R}}$ as already formalized in the project; the second is its reflected (odd-part) companion.

Lemma 4.3 (`weilKernelLog_eq_sinhKernel_add`). *For $u \neq 0$,*

$$\frac{\sqrt{e^u}}{|1 - e^u|} = \frac{1}{2 \sinh(|u|/2)} = \sinh\mathrm{Kernel}(u) + \sinh\mathrm{KernelRef}(u).$$

Lemma 4.3 explains, quantitatively, why the full line is the wrong model: the limiting density on $L^2(\mathbb{R})$ is $\sinh\mathrm{Kernel} + \sinh\mathrm{KernelRef}$, whereas the paper's Sonin picture sees $\sinh\mathrm{Kernel}$ alone. The discrepancy is an explicit, controlled term, denoted $E(F)$ below; Section 6 removes it.

5 The logarithmic profile and the finite part on the full line

5.1 Separation of the divergence

Proposition 5.1 (`RenormalizedFinitePart.lean`). *Let $f \in C_c(\mathbb{R}_+^*, \mathbb{C})$ satisfy $\|f(\lambda)\| \leq C|1 - \lambda|$, i.e. f vanishes at the singularity at a Lipschitz rate. Then*

$$\mathrm{Tr}\left(\vartheta(f)S^{(\Lambda)}\right) \longrightarrow \int_{\mathbb{R}_+^*} f(\lambda) \frac{\sqrt{\lambda}}{|1 - \lambda|} d^*\lambda.$$

No counter-term is needed for such f , so the limit *is* the finite part. The proof is dominated convergence with the cut-off-independent bound of Theorem 4.1 as dominating function. Linearity of the cut-off trace in f then reduces the general case to a single reference function.

5.2 The reference profile

Let $\mathrm{refBump}(\lambda) = \max(0, 1 - |\log \lambda|)$, the triangular bump of unit half-width in the logarithmic coordinate. Writing $T = 2\pi\Lambda^2$ and $\psi(u) = (e^u - 1)/\max(1, e^u)$, the density becomes

$$\kappa_\Lambda(e^u) = \frac{\mathrm{Si}(T\psi(u))}{\pi \sinh(u/2)},$$

so that

$$\theta(T) = \int_{\mathbb{R}} \text{tri}(u) \frac{\text{Si}(T\psi(u))}{\pi \sinh(u/2)} du.$$

The proof splits $1/(\pi \sinh(u/2)) = w(u) + 2/(\pi u)$ into a regular part and the exact hyperbolic singularity, giving $\theta = \theta_A + \theta_B + \theta_C$, where θ_A converges by dominated convergence, $\theta_B \rightarrow 0$ by the uniform estimate $|\text{Si}(T\psi(u)) - \text{Si}(Tu)| \leq 2|u|$ (`CutoffLogProfileAux.lean`), and

$$\theta_C(T) = \frac{4}{\pi} \left(\int_0^T \frac{\text{Si}(s)}{s} ds - \int_0^1 \text{Si}(Tu) du \right)$$

carries the divergence.

Theorem 5.2 (`CutoffLogProfile.lean`, `CutoffConstants.lean`). $\theta(T) - 2 \log T$ converges. Consequently

$$\text{Tr}(\vartheta(\text{refBump})S^{(\Lambda)}) = 4 \log \Lambda + z_0 + o(1),$$

with the explicit value

$$z_0 = z_{\text{Ref}} = \theta_A^{\text{val}} + \frac{4}{\pi} \left(K - \frac{\pi}{2} \right) + 2 \log(2\pi),$$

where

$$K = \int_0^1 \frac{\text{Si}(s)}{s} ds + \int_1^\infty \frac{\text{Si}(s) - \pi/2}{s} ds$$

and $\theta_A^{\text{val}} = \int_{\mathbb{R}} \text{tri}(u) \frac{\pi}{2} \text{sgn}(u) w(u) du$.

The two analytic inputs are $\int_0^T \text{Si}(s) \frac{ds}{s} - \frac{\pi}{2} \log T \rightarrow K$ and $\int_0^1 \text{Si}(Tu) du \rightarrow \pi/2$. The constant K is denoted `siLogConst` in the formalization; Section 7 evaluates it.

5.3 The unconditional full-line finite part

Theorem 5.3 (`CutoffLogProfile.lean`). There is a universal constant z_1 such that for every $f \in C_c(\mathbb{R}_+^*, \mathbb{C})$ which is Lipschitz at $\lambda = 1$, i.e. $\|f(\lambda) - f(1)\| \leq C|1 - \lambda|$,

$$\text{Tr}(\vartheta(f)S^{(\Lambda)}) - 4f(1) \log \Lambda \rightarrow W_{\mathbb{R}}(\Delta^{-1/2}F) + E(F) + f(1) z_1,$$

equivalently, using $W_{\mathbb{R}}(\Delta^{-1/2}F) = D(F) - L_{\text{Norm}}(F)$,

$$\rightarrow D(F) - L_{\text{Norm}}(F) + E(F) + f(1) z_1,$$

where F is the logarithmic transform of f and E is the regularized pairing against `sinhKernelRef`.

In Lean:

```
theorem exists_universal_finitePart_unconditional (b : HilbertBasis iota C L2R) :
  exists z1 : C, forall (f : C_c(Rplus, C)) (C : R),
    (forall lam : Rplus, |f lam - f 1| <= C * |1 - lam|) ->
      Tendsto (fun cut => modelCutTrace b cut f - f 1 * 4 * log cut) atTop
        (nhds (WeilR (deltaHalfInv (ofLog (logTest f)))
          + reflectionPairingReg (logTest f) + f 1 * z1))
```

Two features deserve emphasis. First, the entire f -dependence of the finite part is the archimedean Weil distribution: the residual constant multiplies only the local value $f(1)$. Second, the statement is *unconditional* — the logarithmic-profile hypothesis that an earlier formulation carried as a `Prop` is discharged by Theorem 5.2, not assumed.

The coefficient here is $4f(1)$, not the paper's $2f(1)$. This is a feature of the full-line model, whose phase-space scale is Λ^2 : indeed $4 \log \Lambda = 2 \log(\Lambda^2)$, consistent with $d(\Lambda) = 4\Lambda^2$ from Theorem 3.3. The next section resolves both this and the $E(F)$ discrepancy simultaneously.

6 The even (Sonin) subspace: the main theorem

6.1 The even projection and its density

Let $(R\xi)(x) = \xi(-x)$ be the reflection and $P_{\text{ev}} = (1 + R)/2$ the orthogonal projection onto even functions; both are self-adjoint (`EvenSemiLocalKernel.lean`). The Weil map $w : L^2(\mathbb{R}_+^*, d^*\rho) \rightarrow L^2(\mathbb{R})$, $(w\xi)(x) = (2|x|)^{-1/2}\xi(|x|)$, is an isometry whose range is exactly the even subspace (`EvenPicture.lean`), so the even subspace is the correct home for the paper's picture.

Theorem 6.1 (`EvenSemiLocalKernel.lean`). *The even semi-local trace density $\kappa_\Lambda^{\text{ev}}(a) = \text{Tr}(D_a S^{(\Lambda)} P_{\text{ev}})$ is, for $a \neq 1$,*

$$\kappa_\Lambda^{\text{ev}}(a) = \frac{1}{2}\sqrt{a} \left[\frac{2}{\pi(a-1)} \text{Si}\left(\frac{2\pi(a-1)\Lambda^2}{\max(1,a)}\right) + \frac{2}{\pi(a+1)} \text{Si}\left(\frac{2\pi(a+1)\Lambda^2}{\max(1,a)}\right) \right].$$

The second summand is the same phase integral as in Theorem 4.1 with $a-1$ replaced by $a+1$; it is the contribution of R . Since $a+1$ never vanishes, that summand is bounded uniformly in the cut-off and converges; it therefore carries *no* divergence (`abs_reflDensityVal_le`, `tendsto_reflModelCutTrace`). The halving identity

$$\text{Tr}\left(\vartheta(f) S^{(\Lambda)} P_{\text{ev}}\right) = \frac{1}{2} \left(\text{Tr}\left(\vartheta(f) S^{(\Lambda)}\right) + \text{Tr}\left(\vartheta(f) S^{(\Lambda)} R\right) \right)$$

(`evenModelCutTrace_eq_half`) therefore halves the logarithmic coefficient from $4f(1)$ to $2f(1)$.

6.2 Exact cancellation of the odd part

Lemma 6.2 (`coshKernelLog_eq_sub`). *In the logarithmic coordinate the reflected kernel is $\sqrt{\lambda}/(1+\lambda) = 1/(2 \cosh(u/2))$, and for $u \neq 0$*

$$\frac{1}{2 \cosh(u/2)} = \sinh\text{Kernel}(u) - \sinh\text{KernelRef}(u).$$

Combining Lemmas 4.3 and 6.2:

$$\frac{1}{2} \left(\frac{1}{2 \sinh(|u|/2)} + \frac{1}{2 \cosh(u/2)} \right) = \sinh\text{Kernel}(u).$$

Setting $s = |u|/2$, both sides equal $e^s/(4 \sinh s \cosh s)$; the identity is exact. Hence averaging the full-line kernel with its reflection returns *precisely* the kernel of $W_{\mathbb{R}}$, and the remainder $E(F)$ of Theorem 5.3 cancels identically.

6.3 The renormalized local trace formula on the Sonin subspace

Theorem 6.3 (`EvenCutoffTrace.lean`). *There is a universal constant z_{ev} such that for every $f \in C_c(\mathbb{R}_+^*, \mathbb{C})$ with $\|f(\lambda) - f(1)\| \leq C|1 - \lambda|$,*

$$\boxed{\text{Tr}\left(\vartheta(f) S^{(\Lambda)} P_{\text{ev}}\right) - 2f(1) \log \Lambda \xrightarrow{\Lambda \rightarrow \infty} W_{\mathbb{R}}\left(\Delta^{-1/2} F\right) + f(1) z_{\text{ev}} = D(F) - L_{\text{Norm}}(F) + f(1) z_{\text{ev}}.}$$

The logarithmic coefficient is exactly $2f(1)$, the normalization of [3], and the odd-part remainder is gone. The finite part is the normalized archimedean functional L_{Norm} , up to the single local constant $f(1)z_{\text{ev}}$.

6.4 Evaluation of the constant in the formalization

Theorem 6.4 (`EvenConstant.lean`). *The constant of Theorem 6.3 is*

$$z_{\text{ev}} = \frac{2K}{\pi} - \gamma,$$

with K the sine-integral logarithm constant of Theorem 5.2 and γ the Euler–Mascheroni constant.

The evaluation proceeds by cancelling every elementary contribution against the reflected pairing. The pivotal auxiliary identity is

$$\theta_A^{\text{val}} + \mathcal{J} = 4 \log 2 + 2, \quad \mathcal{J} = \int_0^\infty \frac{1 - \text{tri}(t)}{\sinh(t/2)} dt,$$

proved as `thetaAVal_add_triJVal` by combining the two integrals over $(0, \infty)$ into $\int_0^\infty \left(\frac{1}{\sinh(t/2)} - \frac{2 \text{tri}(t)}{t} \right) dt$. Together with the closed form of $W_{\mathbb{R}}$ on Lipschitz test functions and the value $c_0 = \log(4\pi) + \gamma + \log 2 + \pi/2$ this yields $z_{\text{Ref}} + \mathcal{J} - 2c_0 + \pi = 2z_{\text{ev}}$ and hence Theorem 6.4.

7 The novel result: the finite-part constant vanishes

Theorem 6.4 leaves the trace formula with a residual local term. We now show that this term is absent: the constant is exactly zero. To our knowledge this evaluation is new in the present context, and it is the one mathematical statement in this paper that was obtained *after*, and as a consequence of, the verification pass rather than during proof development.

Lemma 7.1. *Let*

$$K = \int_0^1 \frac{\text{Si}(s)}{s} ds + \int_1^\infty \frac{\text{Si}(s) - \pi/2}{s} ds = \lim_{X \rightarrow \infty} \left(\int_0^X \frac{\text{Si}(s)}{s} ds - \frac{\pi}{2} \log X \right).$$

Then

$$K = \frac{\pi\gamma}{2}.$$

Proof. Integrate by parts on $[0, X]$, using $\text{Si}'(s) = \sin(s)/s$ and $\text{Si}(s) = O(s)$ as $s \rightarrow 0^+$:

$$\int_0^X \frac{\text{Si}(s)}{s} ds = \text{Si}(X) \log X - \int_0^X \frac{\sin s}{s} \log s ds.$$

Since $\text{Si}(X) - \pi/2 = O(1/X)$, we have $\text{Si}(X) \log X - \frac{\pi}{2} \log X \rightarrow 0$, whence

$$K = - \int_0^\infty \frac{\sin s}{s} \log s ds.$$

To evaluate the remaining integral, consider the Mellin-type transform

$$\Phi(a) = \int_0^\infty s^{a-1} \sin s ds = \Gamma(a) \sin\left(\frac{\pi a}{2}\right), \quad 0 < a < 1,$$

a classical evaluation [4, §3.761], valid by analytic continuation in a neighbourhood of $a = 0$ in the sense of the regularized integral. Differentiating under the integral sign at $a = 0$ gives $\Phi'(0) = \int_0^\infty s^{-1} \sin(s) \log s ds$. On the other hand, from the Laurent expansions $\Gamma(a) = a^{-1} - \gamma + O(a)$ and $\sin(\pi a/2) = \frac{\pi a}{2} + O(a^3)$ we obtain

$$\Phi(a) = \left(\frac{1}{a} - \gamma + O(a) \right) \left(\frac{\pi a}{2} + O(a^3) \right) = \frac{\pi}{2} - \frac{\pi\gamma}{2} a + O(a^2),$$

so $\Phi(0) = \pi/2$ — the Dirichlet integral, as a consistency check — and $\Phi'(0) = -\frac{\pi\gamma}{2}$. Therefore

$$\int_0^\infty \frac{\sin s}{s} \log s ds = -\frac{\pi\gamma}{2}, \quad \text{and} \quad K = \frac{\pi\gamma}{2}. \quad \square$$

Theorem 7.2. $z_{\text{ev}} = 0$. Consequently Theorem 6.3 takes the clean form

$$\boxed{\text{Tr}\left(\vartheta(f) S^{(\Lambda)} P_{\text{ev}}\right) - 2f(1) \log \Lambda \xrightarrow{\Lambda \rightarrow \infty} W_{\mathbb{R}}\left(\Delta^{-1/2} F\right) = D(F) - L_{\text{Norm}}(F),}$$

with no local correction term.

Proof. By Theorem 6.4 and Lemma 7.1,

$$z_{\text{ev}} = \frac{2K}{\pi} - \gamma = \frac{2}{\pi} \cdot \frac{\pi\gamma}{2} - \gamma = \gamma - \gamma = 0. \quad \square$$

Remark 7.3. The vanishing is not a coincidence of normalization. The constant $c_0 = \log(4\pi) + \gamma + \log 2 + \pi/2$ in the closed form of $W_{\mathbb{R}}(\Delta^{-1/2} \cdot)$ contains γ , and the sine-integral constant K produced by the cut-off contains the *same* γ with the reciprocal weight $2/\pi$. That the two cancel exactly is precisely the statement that the sharp phase-space cut-off is a *normalization-free* regularization of the archimedean local term: the only price of the cut-off is the $2f(1) \log \Lambda$ counter-term, whose coefficient is itself canonical (`RenormalizedTraceUniqueness.lean`).

Remark 7.4 (Status in Lean). Theorem 6.4 is machine-checked. Lemma 7.1 and Theorem 7.2 are *not yet* formalized: what is needed is the single Lean lemma `siLogConst =  $\pi * \text{eulerMascheroniConstant} / 2$` . It is self-contained and, given the Mathlib API for Γ and for Si , of modest size; we flag it as the immediate next target. Until it is discharged, Theorem 7.2 should be read as a classical (hand-proved) corollary of a machine-checked theorem, and is labelled as such throughout.

8 Independent numerical corroboration

Because the constants above pass through several cancellations, we reproduced them numerically *ab initio* from the Lean definitions — `logDens`, `psiCut`, `wKer`, `triReal` — in an independent computer-algebra environment, without reference to the Lean proofs. This is a check of the *statements*, complementary to the kernel’s check of the *proofs*.

The logarithmic profile. Direct quadrature of $\theta(T) = \int \text{tri}(u) \text{Si}(T\psi(u)) / (\pi \sinh(u/2)) du$ gives:

T	$\theta(T) - 2 \log T$
10^2	−0.85295856
10^3	−0.85870093
10^4	−0.85927387
10^5	−0.85933117
10^6	−0.85933694
predicted $z_{\text{Ref}} - 2 \log(2\pi)$	−0.85933753

The predicted value uses $\theta_A^{\text{val}} + \frac{4}{\pi}(K - \frac{\pi}{2})$ with $K = \pi\gamma/2$ from Lemma 7.1. Agreement is to 6×10^{-7} , within quadrature error. Note that this simultaneously corroborates Theorem 5.2 and Lemma 7.1: using an imprecise numerical value of K instead of the closed form degrades the agreement by three orders of magnitude.

The elementary cancellation. $\theta_A^{\text{val}} + \mathcal{J} = 4 \log 2 + 2$: numerically $-0.0137689 + 4.786358 = 4.772589$ against $4 \log 2 + 2 = 4.772589$.

The constant K . Both the defining double integral and the closed form $\pi\gamma/2$ evaluate to 0.906688246195802 to fifteen significant figures.

The final constant. $z_{\text{Ref}} + \mathcal{J} - 2c_0 + \pi = 3.2 \times 10^{-6}$, consistent with $2z_{\text{ev}} = 0$ at the precision of the quadrature.

9 The formalization

9.1 Environment and scale

The development targets Lean 4 (toolchain `leanprover/lean4:v4.28.0`) against Mathlib at revision `v4.28.0` [6, 7]. It comprises 305 modules and approximately 2.2×10^5 lines. Table 1 lists the modules of the operator-theoretic chain.

Table 1: Modules of the renormalized-trace development (lines of Lean).

Module	Content	Lines
Dilation	concrete dilation unitary, intertwining with ϑ	239
HilbertSchmidtPairing	HS inner product as a kernel integral	147
SemiLocalKernel	$\kappa_\Lambda(a) = \text{Tr}(D_a S^{(\Lambda)})$, factorization	266
SemiLocalSi	closed form, uniform bound, off-diagonal limit	338
DiagonalExact	$d(\Lambda) = 4\Lambda^2$	197
DiagonalQuadraticGrowth	refutation of the diagonal-log hypothesis	574
RenormalizedFinitePart	finite part for functions vanishing at 1	445
WeilKernelComparison	kernel splitting, identification with $W_{\mathbb{R}}$	341
RenormalizedAsymptotics	universal finite part, conditional form	254
CutoffLogProfileAux	elementary Si and compression estimates	233
CutoffLogProfile	the logarithmic profile; unconditional finite part	736
CutoffConstants	z_{Ref} , <code>siLogConst</code>	199
EvenSemiLocalKernel	P_{ev} , even density in closed form	419
EvenCutoffTrace	halving identity; main theorem	529
EvenConstantAux	$\theta_A^{\text{val}} + \mathcal{J} = 4 \log 2 + 2$	270
EvenConstant	evaluation of z_{ev}	215
EvenPicture	the Weil map and the even subspace	471
RenormalizedTrace(+Density, Uniqueness)	abstract renormalized trace, canonicity	632
SemiLocalDiagonal, NearDiagonalKernel	diagonal machinery	913
TraceIdentityObstruction	refutation at fixed cut-off	380
UnnormalizedCounterexample	refutation of unnormalized positivity	677

9.2 Soundness discipline

Three disciplines were applied throughout and are worth recording, since they are what makes a formalization of this size auditable.

1. **No axioms.** The project declares no axioms of its own. Analytic inputs that are not proved are carried as `Prop`-valued hypotheses of the theorems that use them, never as axioms, so that a vacuous or false hypothesis cannot silently propagate. `AxiomAudit.lean` contains 62 `#print axioms` commands, re-elaborated on every build; each reports exactly `[propxt, Classical.choice, Quot.sound]`.
2. **Hypotheses are discharged, not accumulated.** The logarithmic-profile hypothesis `HasCutoffLogProfile` was introduced in one run as a `Prop` and *discharged* in a later one by `hasCutoffLogProfile_refBump`, with an explicit constant. The finite-part theorems

of Sections 5–6 invoke the discharged form. A hypothesis-carrying theorem whose hypothesis is never satisfied is vacuously true and worthless; witnesses are supplied for every hypothesis class used (e.g. `refBump_lipschitz` for the Lipschitz condition).

3. **Refutations are proved, not conjectured.** Where a plausible route fails, the failure is formalized as a negation (Theorems 3.1, 3.2, 3.3), which is strictly more informative than abandoning the route.

A whole-project search confirms that the token `sorry` occurs only inside comment blocks: the transcription of the original informal skeleton in `Skeleton.lean`, two withdrawn statements recorded for the record in `FourierSide.lean`, and the withdrawn `L_positive` in `Positivity.lean` (withdrawn because it is *false*, by Theorem 3.2).

10 Limitations and open problems

We state the boundary of the result precisely.

(L1) Model versus abstract trace functional. The objects `modelCutTrace` and `evenModelCutTrace` are *defined* as the kernel integrals $\int f(\lambda)\kappa_\Lambda(\lambda^{-1})d^*\lambda$, with κ_Λ a genuine operator trace. What is not yet formalized is the identity between these and the abstract functional `cutTrace b U  $\Lambda$  f` parameterized by an arbitrary unitary identification U of the two pictures, in terms of which the project’s statement `RenormalizedTraceIdentity` is phrased. On the full line the obstruction was real — no such U exists, since the Weil map is an isometry onto the even subspace only. On the even subspace, however, that obstruction disappears: `weilUnitary` is exactly the required U . Composing (i) `weilMap_scaling_eq_dil` with (ii) an even analogue of the trace/Bochner interchange `cutTrace_eq_integral` would close the gap, and together with Theorem 7.2 would yield `RenormalizedTraceIdentity weilUnitary` outright. We regard this as the single most valuable remaining step.

(L2) The full-line constant z_1 . The constant of Theorem 5.3 exists unconditionally but is not evaluated in closed form. It is not needed for the even statement, where z_{ev} is evaluated and, by Theorem 7.2, vanishes.

(L3) Formalization of Lemma 7.1. As noted, `siLogConst` = $\pi\gamma/2$ is proved here classically and not yet in Lean.

(L4) Scope. The development treats the archimedean place. Nothing here addresses the finite places, the global explicit formula, or Weil positivity as a whole; in particular no claim whatsoever is made regarding the Riemann hypothesis.

11 Methodological note: adversarial verification in a multi-agent workflow

The development was produced across a sequence of autonomous proof-development runs, each delivering a source archive and a natural-language summary, with an independent verification pass applied to every delivered archive before its claims were accepted.

That pass proved to be load-bearing. In one instance the summary described three modules (`RenormalizedFinitePart`, `WeilKernelComparison`, `RenormalizedAsymptotics`) with named theorems; ten of the eleven identifiers named did not occur anywhere in the delivered archive, which was in fact an earlier snapshot. The discrepancy was mechanical to detect and would

have been invisible to a reader of the summary alone. In a second instance the verification pass identified that the residual constant z_{ev} , reported as an irreducible datum, had a classical closed form making it zero — Theorem 7.2, the paper’s new result.

Two practices generalize. First, *never accept a claim about an artifact without inspecting the artifact*: check that named declarations exist, that their statements match the prose, and that no hypothesis is doing the work of a **sorry**. Second, *reproduce the constants independently*: a formalization can be entirely correct and still leave a quantity in a form that conceals its value, and numerical reconstruction from the definitions is cheap, orthogonal to the proof kernel, and — as here — occasionally productive of new mathematics.

12 Conclusion

We have presented a **sorry**-free Lean 4 formalization establishing, in a concrete model on $L^2(\mathbb{R})$, a renormalized local trace formula at the archimedean place: the sharp phase-space cut-off $S^{(\Lambda)}$ has semi-local density expressible exactly through the sine integral, the density converges off-diagonal to the archimedean Weil kernel $\lambda^{1/2}/|1 - \lambda|$, the cut-off trace diverges logarithmically with a canonical coefficient, and on the even (Sonin) subspace the renormalized limit is the normalized archimedean functional L_{Norm} with the paper’s coefficient $2f(1)$. Three plausible alternative formulations were refuted by machine-checked counterexamples.

The new mathematical contribution is the evaluation $K = \pi\gamma/2$ of the sine-integral logarithm constant, whence the finite-part constant z_{ev} vanishes identically and the renormalized trace formula on the Sonin subspace holds with no local correction:

$$\text{Tr}\left(\vartheta(f)S^{(\Lambda)}P_{\text{ev}}\right) = 2f(1)\log\Lambda + D(F) - L_{\text{Norm}}(F) + o(1).$$

Availability

The Lean sources, the module inventory, the axiom audit, and the verification notes are maintained in the DULA **prime-inertia-engine** repository.

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